Virtual Mouse Control Using Hand Gesture Recognition and Voice Assistant

Nitheesh Shetty
Dept. of Computer Science and Engineering
R N S Institute of Technology
Bengaluru, India
1rn19cs091.nitheeshshetty@gmail.com

Nitish. K
Dept. of Computer Science and Engineering
R N S Institute of Technology
Bengaluru, India
1rn19cs092.nitish@gmail.com

Rakshith B K
Dept. of Computer Science and Engineering
R N S Institute of Technology
Bengaluru, India
1rn19cs108.rakshith@gmail.com

Rajatha Bangera
Dept. of Computer Science and Engineering
R N S Institute of Technology
Bengaluru, India
1rn20cs411.rajath@gmail.com

Yashasvi B N
Dept. of Computer Science and Engineering
R N S Institute of Technology
Bengaluru, India
yashasvibn@rnsit.ac.in

Abstract - The virtual mouse is a computer interface that uses hand gesture recognition and voice control to manipulate the cursor on the screen. This paper proposes a system for virtual mouse control using hand gestures and voice commands .The Virtual mouse using hand gesture recognition is a computer interface platform where the desktop cursor is being moved by the hand rather than using the mouse for the control movement for the cursor. It detects the points fed to the system of the palm as input and analyses the movement of the palm and does the functioning based on the given input points of the palm. The voice control uses Natural Language Processing techniques to interrupt the voice commands and performs the necessary actions. The proposed system was implemented and tested in the desktop computers and the Laptops. The result shows that the system is effective and can be used as an alternate to the mouse. This proposed system can be a benefit to individuals with physical disabilities , as well as to improve the upcoming generations to the most digital world.

Keywords—Voice Assistance, Voice recognition, Basic operations, Web-cam, Hand Recognition, Palm detection, Gesture recognition, , Python , MediaPipe, OpenCV, NumPy

I. INTRODUCTION

The present computer mouse has been the input for the movement of cursor from many decades which might not be helpful for the individuals with the physical abilities. Additionally, using mouse for long hours can also cause stress and might lead to aggravation of the user's in some point of time.

To address these limitations, the researcher's and developers have introduced Touchscreen's, Trackpads and voice control etc., We have come up with another way of eliminating the using of mouse by virtually controlling the mouse by the movement of our hands and voice control.

This paper proposes a system of virtual mouse using hand gesture recognition and voice assistance to control the movement of the cursor on the Screen. By combining these technologies the proposed system offers an alternative input method that is more natural, intuitive and potentially accessible than traditional mouse devices.

Project essential feature:

- User Friendly
- Portable
- Handle simple operations like clicking, drag-drop, copy-paste etc.,

II. EXISTING SYSTEM

The existing system consists of a wired or wireless mouses for the movement of the cursor on the screen. To eleminate this, in the early times researcher's and developers developed the system called the Virtual mouse using Hand Gesture Recognization. In the early systems the proposed system might function for particular set of input images i.e. it might function only for the scenarios where the user's hand/palm image matches the trained set of images given to the System. It might not work for any extra actions performed by the user. In the previosuly proposed/existing systems only the hand gesture feature was being implemented, so it might lead difficulties for the individuals with physical dissabilities.

Comparing to the previously acquired technologies to build the system, the system used to recognize the hand, finger tips, points etc., was highly complex to understand and difficult to use

III. PROPOSED SYSTEM

- Using the current system even-though there are a number of quick access methods available for the hand and mouse gesture for the laptops, using our project we could make use of the laptop or web-cam and by recognizing the hand gesture
- We could control mouse and perform basic operations like mouse pointer controlling, select and deselect using left click, and a quick access feature for file transfer
- We have mainly concentrated on the mouse pointing and clicking actions along with an action for the file transfer between folders by hand action and the movements. The

system we are implementing which is been written in python code

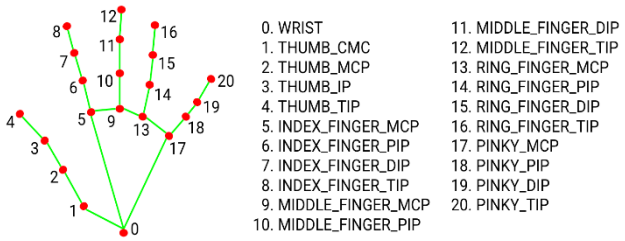


Figure 1.Palm Detection Model

- In this proposed system, we haven't fed any example inputs of the hand movement and trained the model, instead we are working on the idea of palm detection model
- Here, it tales all the given points shown in the above as the input. As the camera is turned on the model works on the basis of the points given to the corresponding user's hand.
- Based on the points fed the movement is made

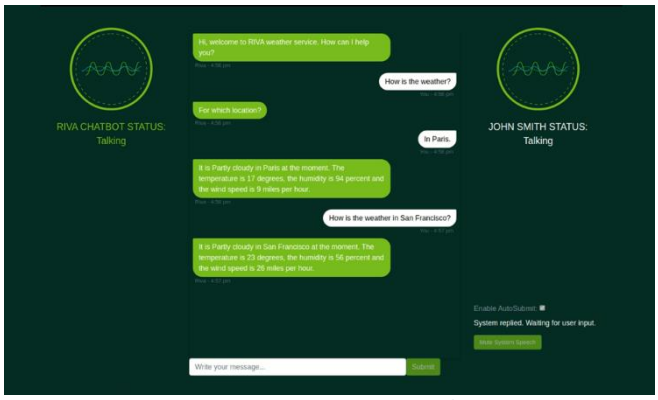


Figure 2.Voice assistance demo fig

- The voice assistance works on the basis of the voice input.
- The voice control uses the Natural Language Processing techniques to interrupt the voice commands and follow the necessary actions
- Commands are given to the proposed system to get any particular work done just by talking to the system.

IV. Techniques used

A. ***BACKGROUND SUBTRACTION METHOD(BSM)*** – It is a computer vision algorithm that detects objects in video content by comparing them to the background and foreground parts of an image.

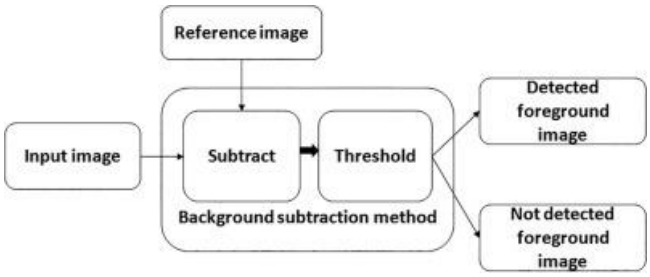


Figure 3. Background Subtraction Method

B. ***SKIN COLOR DETECTION*** – Process of finding skin colored pixels and regions in an image or a video

C. ***CONTOUR ANALYSIS*** – To identify the hand and extract its features. It is used to find the finger tips and valley points, segmenting the skin color and conversion of binary image.

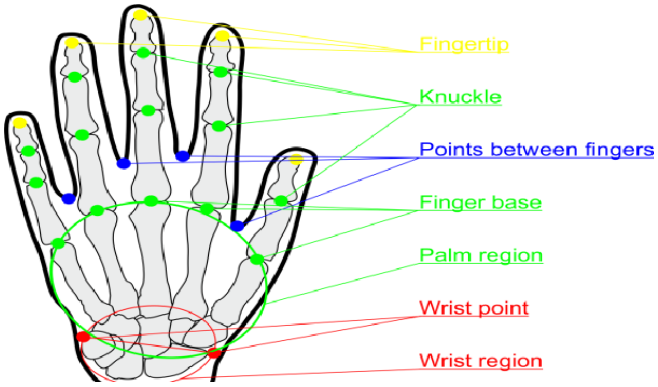


Figure 4. Contour Analysis

The extracted feature from this technique is used to recognize different hand and extract its features

V. Flowchart

A. *FOR HAND GESTURE AND MOVEMENT RECOGNITION:*

[1] Turn on the web cam.

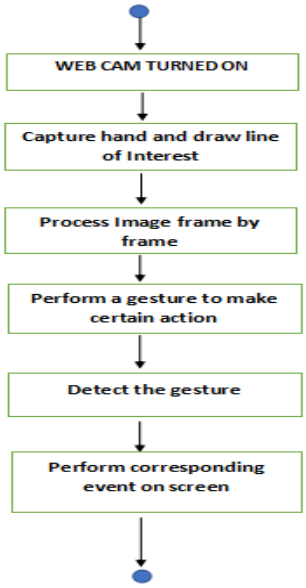
[2] Capture the image as a frame.

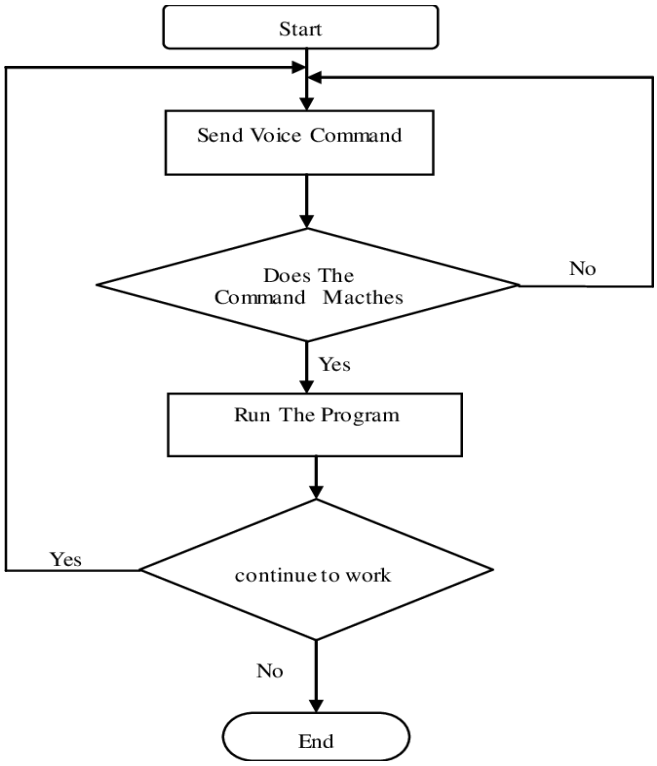
[3] Process the frame captured by the web cam, after processing convert the image HSV to RGB format

[4] Perform any gestures by the hand to perform any specific action

[5] Detect the gestures made by using contour analysis method

[6] Perform corresponding event on the screen





B. FOR VOICE CONTROL ASSISTANCE:

[1] Start the Voice assistant prototype

[2] Send Voice commands to the Voice assistant i.e. Proton in our proposed system

[3] If the command is clear and matches means, it is ready to launch the Virtual Cursor prototype and also can perform other actions as well if the command is valid

[4] If the command does not matches means the voice command should be sent again to the voice assistant prototype

[5] End the program as soon as the executable actions finishes

VI. METHODOLOGY

Following steps are included to run this prototype:

[1] The first step is to run the voice assistant prototype As shown in the figure 5.

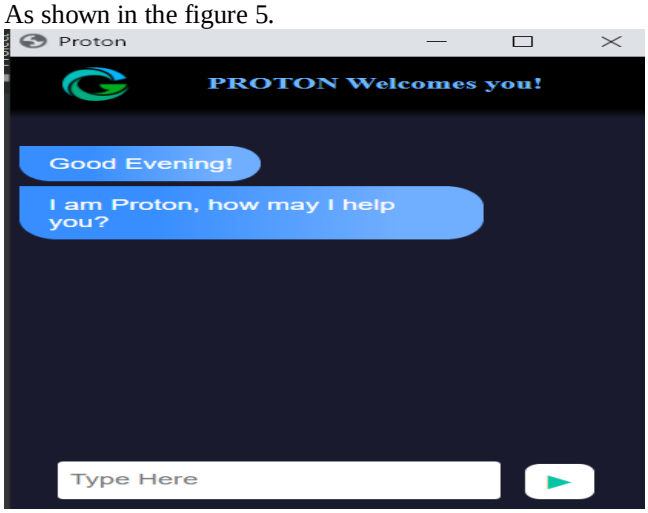


Figure 5. Beginning of Proton(Wishing the user based upon the current time)

[2] The second step is to talk with the voice assistant(proton) and can also make it to perform some basic actions as shown in the Figure 6, 7, 8, 9 and 10

Figure 6. Performing Date and Time operation with Proton

Figure 7. Listing the files present in windows with Proton

Figure 8. Exit operation to stop the program running

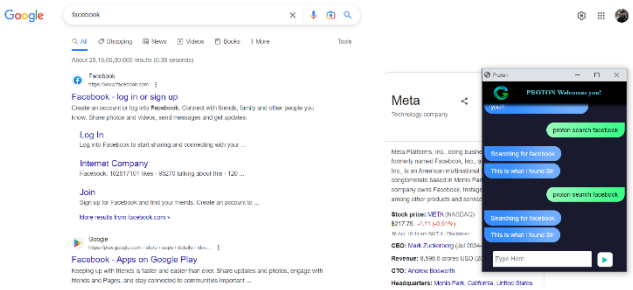


Figure 9. Performing search operations by giving voice command to Proton

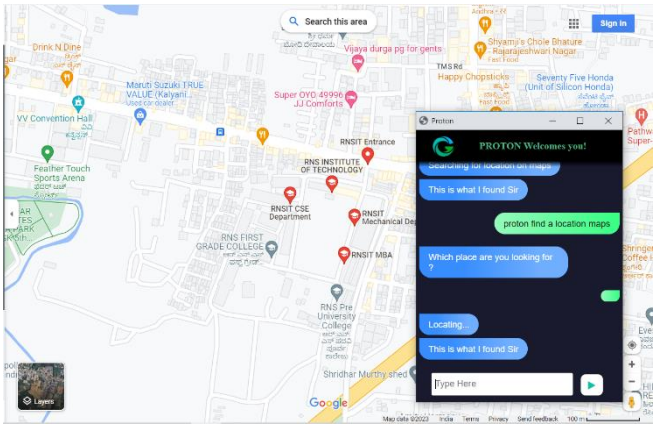


Figure 10. Searching location using Proton on Google Maps by providing voice commands

[3] The third step is to say to the proton to launch the Virtual Cursor Prototype. As soon as the proton gets the input it launches the Virtual Cursor Programs as shown in the Figure 11.

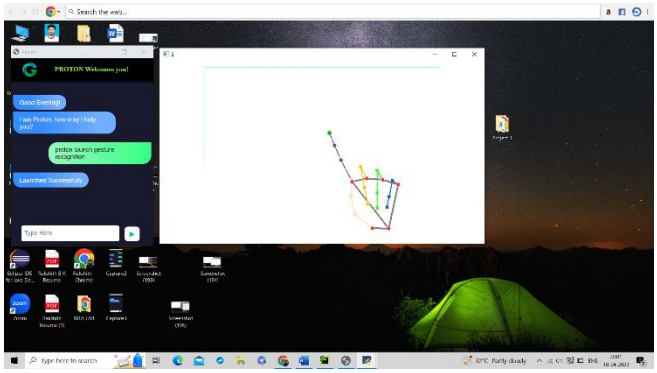


Figure 11. Launching Hand Gesture Recognition by giving command to Proton

[4] In the next step, as soon as the proton launches the Virtual Cursor Program, the system automatically turns the camera on

[5] In this step, The camera captures the required inputs i.e. the user needs to place the palm in front of the camera and should perform the required necessary actions to produce the output

[6] Index finger is used for moving the cursor as soon as the camera turns on as shown in the figure 12.

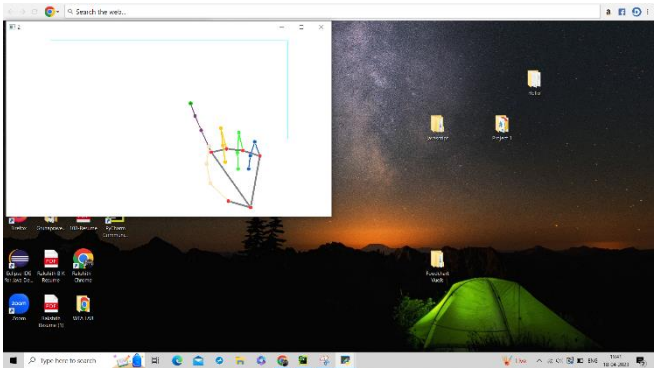


Figure 12. Moving Cursor with the help of Index Finger

[7] For clicking options, the user needs to touch with the index finger with the thumb finger and the index finger with the middle finger for right and left clicks
[8] To perform the right click operation the user needs to join the index finger with the thumb finger as shown in the Figure 13.

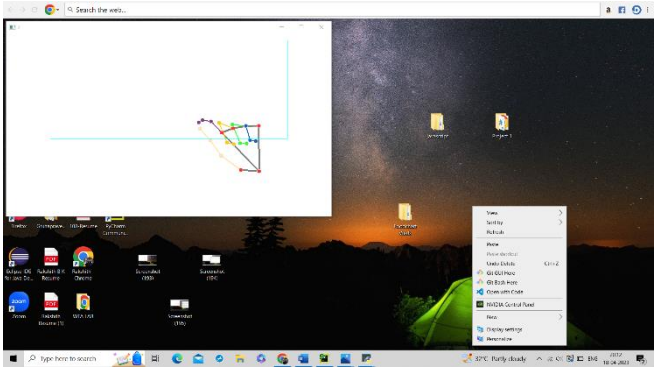


Figure 13. Right Click operation

[9] For performing the left click operation the user needs to join the index finger with the middle finger as shown in the figure 14.

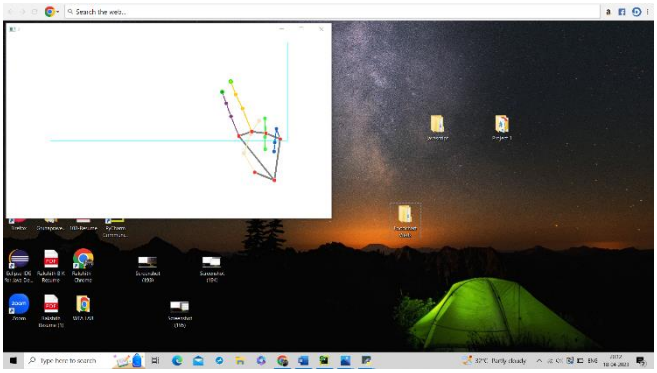


Figure 14. Left Click operation

[10] For the drag and drop operation the user needs to combine three fingers starting from the index finger and can perform the drag and drop operation as shown in the figure 15.

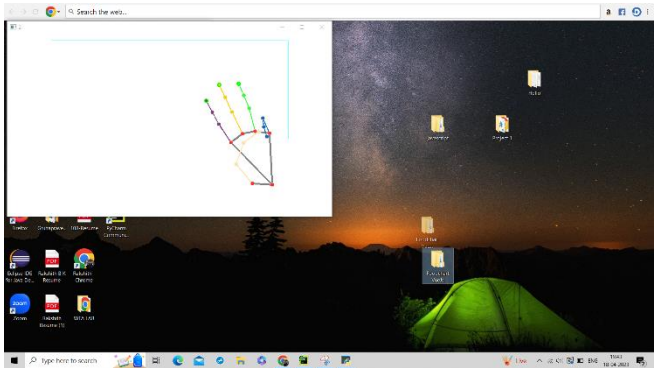


Figure 15. Drag and Drop

[11] For selecting any texts in the paragraph we can use the same as the above method like combining the three fingers starting from the index finger as shown in the figure 16.

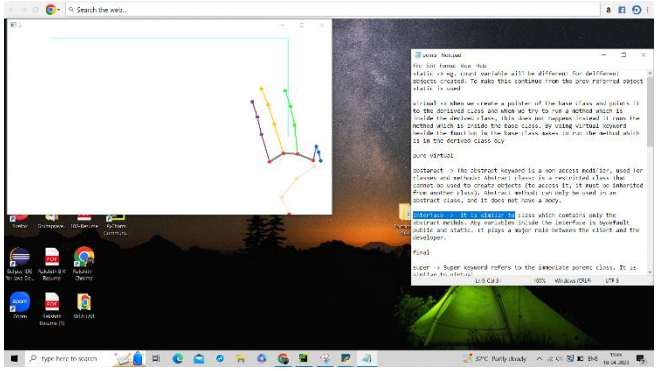


Figure 16. Selecting Texts

[12] All these operations of the fingers works on the basis of points fed to the model of the palm
[13] Each point of the finger is calculated and integrated to perform different kind of actions

VII. Results and Evaluation

The motive of this paper was to make the machine more interactive and responsive towards human behavior. The sole aim of this paper was to make a technology that is affordable and portable with any standard operating system.

The proposed system is used to control the pointer of the mouse by detecting the human hand and moving the pointer in the direction in the human hand respectively. the system Control simple function of the mouse such as left-clicking, dragging and cursor movement.

The proposed system may also be controlled by using the voice assistance called Proton. The voice assistance may also be used to perform some simple operations. To launch the virtual cursor prototype by using the voice assistance just say “Proton launch Virtual Cursor”.

Here are some possible results and evaluation metrics that could be used to explain the project in a more detailed way

[1] Gesture recognition accuracy: One important metric for the evaluating the performance of the system would be the accuracy of the gesture recognition algorithm. This could be

measured by comparing the systems detected hand gestures against a ground truth dataset or through user testing

[2] Cursor movement precision: Another important metric can be the precision of the cursor movement, as this is critical for accurate interaction with the user interface. This could be measured by comparing the cursor movements made by the user with the actual movements of the cursor on the screen

[3] Voice command recognition accuracy: The accuracy of the voice command recognition system could be evaluated by the system's transcription of the spoken commands with the actual spoken commands.

[4] System Performance: The system's performance could be evaluated by measuring its resource usage(CPU and memory usage), as well as it's responsiveness and latency during use.

Sl. No	**Comparison of existing and Proposed System**		
	Feature	**Existing System**	**Proposed System**
1)	Stability	The existing System is poor in stability due to usage of old models and techniques	The proposed System is very stable compared to its predecessor
2)	Environment	The existing System is highly dependent on environment in which the system is being used	The proposed system has the very less dependency on the environmental factors
3)	Input modalities	Existing System for virtual cursor control may rely on traditional input devices such as mouse, touchpad etc.,	The proposed system uses hand gestures and voice recognitions as the input modalities
4)	Voice Control Assistance	Existing Virtual cursor systems may has the limited integration options	The proposed system may integrate with other applications or assistive technologies to provide additional functionality such as speech recognition.
5)	Accessibility	It has less accessibility features. It is suitable for the older generations only	The proposed system has improves accessibility features compared to existing systems, as it relies on natural input modalities that may be easier for some users with disabilities to use

VIII. APPLICATION OF PROPOSED WORK

The proposed work can easily be placed as an alternative for the traditional mouse system that has been in existence. With the help of this system, sometimes when the hardware device mouse isn't easily available then at that time user can easily manage the movements of the cursor with the help of showing hand gestures or can use the voice assistant prototype to perform any kind of simple operations.

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